



CIE - II - SOLUTIONS

Topics: Linear Programming & Nonlinear Constrained Optimization

UMAT-503: Optimization Techniques

Instructor: **Dr. D Bhanu Prakash**

Date: 09-09-2026

Q1. Consider the optimization problem:

Maximize $z = 3x_1 + 5x_2$ s.t. $x_1 \leq 4$, $2x_2 \leq 12$, $3x_1 + 2x_2 \leq 18$, $x_1, x_2 \geq 0$

a. Solve by the simplex method. [Marks: 6 | BTL-3]

Solution: Adding slacks s_1, s_2, s_3 : $z - 3x_1 - 5x_2 = 0$; $x_1 + s_1 = 4$; $2x_2 + s_2 = 12$;
 $3x_1 + 2x_2 + s_3 = 18$.

Tableau 1 (basis s_1, s_2, s_3):

Basis	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
z	-3	-5	0	0	0	0	—
s_1	1	0	1	0	0	4	—
s_2	0	2	0	1	0	12	$6 \leftarrow$
s_3	3	2	0	0	1	18	9

Entering x_2 , leaving s_2 ; pivot = 2.

Tableau 2 (basis s_1, x_2, s_3):

Basis	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
z	-3	0	0	2.5	0	30	—
s_1	1	0	1	0	0	4	4
x_2	0	1	0	0.5	0	6	—
s_3	3	0	0	-1	1	6	$2 \leftarrow$

Entering x_1 , leaving s_3 ; pivot = 3.

Tableau 3 (basis s_1, x_2, x_1):

Basis	x_1	x_2	s_1	s_2	s_3	RHS
z	0	0	0	1.5	1	36
s_1	0	0	1	1/3	-1/3	2
x_2	0	1	0	1/2	0	6
x_1	1	0	0	-1/3	1/3	2

All reduced costs in row z are ≥ 0 , so the tableau is **optimal**:

$$x^* = (2, 6)^T, \quad z^* = 3(2) + 5(6) = 36, \quad s_1 = 2 \text{ (constraint 1 is slack)}$$

b. Write the dual for the given problem. [Marks: 3 | BTL-3]

Solution:

Minimize $w = 4y_1 + 12y_2 + 18y_3$ s.t. $y_1 + 3y_3 \geq 3$, $2y_2 + 2y_3 \geq 5$, $y_1, y_2, y_3 \geq 0$

c. Read its optimal solution from the final tableau; [Marks: 3 | BTL-4]

Solution: The optimal dual variables are the z-row coefficients under the slack columns:

$$y^* = (y_1, y_2, y_3) = (0, 1.5, 1)$$

d. Verify strong duality. [Marks: 3 | BTL-4]

Solution: Feasibility check: $0 + 3(1) = 3 \geq 3$ ✓; $2(1.5) + 2(1) = 5 \geq 5$ ✓.

$$w^* = 4(0) + 12(1.5) + 18(1) = 18 + 18 = 36 = z^* \text{ ✓ (strong duality)}$$

Complementary slackness is consistent: $s_1 = 2 > 0 \Rightarrow y_1 = 0$, and $y_2, y_3 > 0 \Rightarrow$ constraints 2 and 3 are binding.

Q2. Minimize $f(x) = x_1^2 + x_2^2 + x_3^2$ subject to

$$h_1(x) = x_1 + x_2 + x_3 - 3 = 0, \quad h_2(x) = x_1 - x_2 + 2x_3 - 4 = 0$$

Apply the Lagrange condition to find the candidate point and verify the second-order sufficiency condition. [Marks: 4+2 | BTL-3]

Solution: **Regularity:** $\nabla h_1 = (1, 1, 1)^T$, $\nabla h_2 = (1, -1, 2)^T$ are linearly independent, so every feasible point is regular.

Lagrange condition: $\nabla f + \lambda_1 \nabla h_1 + \lambda_2 \nabla h_2 = 0$:

$$2x_1 + \lambda_1 + \lambda_2 = 0, \quad 2x_2 + \lambda_1 - \lambda_2 = 0, \quad 2x_3 + \lambda_1 + 2\lambda_2 = 0$$

Put $u = -\lambda_1/2$, $v = -\lambda_2/2$, so that $x = u(1, 1, 1)^T + v(1, -1, 2)^T$. Substituting into the constraints:

$$3u + 2v = 3, \quad 2u + 6v = 4$$

Solving: $v = \frac{3}{7}$, $u = \frac{5}{7}$, hence

$$x^* = \left(\frac{8}{7}, \frac{2}{7}, \frac{11}{7}\right)^T, \quad \lambda_1 = -\frac{10}{7}, \quad \lambda_2 = -\frac{6}{7}$$

Check: $\frac{8+2+11}{7} = 3 \checkmark$ and $\frac{8-2+22}{7} = 4 \checkmark$, with $f(x^*) = \frac{64+4+121}{49} = \frac{27}{7} \approx 3.857$.

Second-order condition: the Lagrangian Hessian is

$$L(x^*, \lambda) = F(x^*) + \lambda_1 \nabla^2 h_1 + \lambda_2 \nabla^2 h_2 = 2I_3$$

For every $y \neq 0$ in the tangent space $T = \{y: \nabla h_1^T y = 0, \nabla h_2^T y = 0\}$ we have

$y^T L y = 2 \|y\|^2 > 0$. Since $L \succ 0$ on T , x^* is a **strict local minimizer** (and, the problem being convex with affine constraints, the global minimizer).

Q3. Answer the following.

a. Define a regular point. [Marks: 2 | BTL-1]

Solution: All columns of gradient of equality constraints are linearly independent.

b. Explain how the Lagrangian method handles irregular points. [Marks: 2 | BTL-2]

Solution: Not handles

- c. The dual of a maximization problem with all $\leq$ constraints is a _____ problem with all _____ constraints. [Marks: 2 | BTL-2]

Solution: minimization; $\geq$

- d. Does every linear programming problem possess a finite optimal solution? Justify your answer. [Marks: 2 | BTL-2]

Solution: False (may be infeasible or unbounded)